AU - Grade 7 - Lines, Angles, and Parallel Line Properti...

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1 What term describes three or more points that all lie on the exact same straight line?

A Concurrent points
B Parallel points
C Collinear points
D Coplanar points

2 When three or more straight lines intersect at a single common point, how are these lines described?

A Perpendicular lines
B Concurrent lines
C Parallel lines
D Collinear lines

3 straight geometric figure starts at a fixed endpoint A and extends infinitely through point B in one direction. How is this figure named?

A Line segment AB
B Ray AB
C Line AB
D Angle AB

4 An angle measures 215°. How is this angle classified?

A Reflex angle
B Revolution
C Acute angle
D Obtuse angle

5 What is the exact angle measure of one complete revolution?

A 360°
B 180°
C 90°
D 270°

6 Two angles are complementary. If one angle measures 34°, what is the measure of the other angle?

A 124°
B 146°
C 56°
D 66°

7 Two adjacent angles lie on a straight line and are supplementary. If one angle is 112°, find the size of the other angle.

A 68°
B 78°
C 158°
D 248°

8 Two straight lines intersect to form opposite angles. If one angle is 73°, what is the size of the vertically opposite angle?

A 107°
B 73°
C 287°
D 173°

9 Four angles meet at a single vertex point. Three of the angles measure 85°, 110°, and 95°. Find the size of the fourth missing angle x.

A 70°
B 60°
C 80°
D 90°

10 Line AB is perpendicular to line CD, intersecting at point O to form right angle AOC (90°). Ray OE is drawn inside angle AOC such that angle AOE is 28°. What is the measure of angle EOC?

A 152°
B 72°
C 62°
D 52°

11 Two parallel lines are cut by a transversal line. If a given angle is 65°, what is the measure of its corresponding angle?

A 180°
B 65°
C 115°
D 25°

12 Two parallel lines are intersected by a transversal. One of the alternate interior angles measures 124°. What is the measure of the other alternate interior angle?

A 66°
B 124°
C 56°
D 180°

13 Two parallel lines are intersected by a transversal. If one co-interior angle is 78°, what is the measure of the other co-interior angle on the same side of the transversal?

A 12°
B 102°
C 112°
D 78°

14 Two straight lines are cut by a transversal line. The interior angles on the same side of the transversal measure 82° and 98°. Are the two lines parallel?

A No, because the two interior angles are not equal
B Yes, because the co-interior angles add up to 180°
C Yes, because alternate angles are equal
D No, because their sum is less than 360°

15 Two adjacent angles on a straight line are given by the expressions (3x + 15)° and (2x + 10)°. Find the value of x.

A 35
B 27
C 25
D 31

16 Two intersecting lines form vertically opposite angles measuring (4x - 20)° and (2x + 30)°. Find the value of x.

A 20
B 30
C 25
D 15

17 Line AB is parallel to line CD. A triangle EFG has its top vertex E on line AB and its base vertices F and G on line CD. If angle AEF = 50° and angle BEG = 60°, calculate interior top angle FEG of the triangle.

A 110°
B 70°
C 80°
D 50°

18 Line L1 is parallel to line L2. A bent path forms an interior vertex V between the lines. The upper angle between L1 and the top arm is 35°, and the lower angle between L2 and the bottom arm is 45°. What is the total measure of interior angle V facin

A 70°
B 100°
C 80°
D 90°

19 Line PQ is parallel to line RS. A transversal intersects PQ at point X and RS at point Y. If interior angle PXY is 135°, what is the size of its co-interior angle XYS on line RS?

A 90°
B 45°
C 180°
D 135°

20 Line m and line n are cut by a transversal line t. The alternate interior angles are measured as 115° and 112°. Which conclusion is correct?

A The lines are perpendicular because they intersect transversal t
B The lines are parallel because both angles are obtuse
C The lines are not parallel because alternate interior angles are not equal
D The lines are parallel because the sum is greater than 180°